

Effect of Dual-rate Sampling on the Stability of a Haptic Interface

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Abstract Conventional controllers in impedance-based haptic interfaces are sampled data systems that utilize position and velocity information for the necessary force feedback. In a canonical virtual wall simulation, this feedback force is generated based on interaction with the wall simulating certain stiffness and damping. Increasing the sampling rate of the controller increases the stable range of virtual wall stiffness. However, an increased sampling rate exacerbates the velocity information, decreasing the stable range of virtual wall damping. In this work, the authors propose a dual-rate sampling scheme in which the position and velocity loops of a haptics controller are decoupled and sampled at different rates. The scheme enables independent sampling of the position data at higher rates, while simultaneously sampling the velocity data at appropriate rates. In this paper, the authors provide experimental and theoretical implications of the effect of dual-rate sampling on the stability of a haptic interface. Experiments with a single degree-of-freedom (DOF) haptic interface reveals

an enhanced region of stable virtual wall stiffness for a particular range of virtual wall damping, compared to the values with conventional uniform-rate scheme. Virtual wall stiffness ranging from 150 - 360 Nm/rad was stably implemented over a range of 0 - 1 Nms/rad of virtual wall damping using the proposed scheme at position loop sampling rate of 20kHz and velocity loop sampling rate of 2kHz. Whereas in the conventional scheme, the stable range of virtual stiffness dropped considerably (~ 0 Nm/rad) for the virtual wall damping above 0.1 Nms/rad when the uniform rate sampling of 20kHz was used for both the position and velocity sampling loops. Theoretical stability analyzes using classical control tools and simulations justified the effectiveness of the proposed scheme. The scheme is easy to implement and extensible to multi-DOF haptic interfaces as well.

Keywords Haptics · Dual-rate sampling · *Z-width*

1 Introduction

Haptics refers to the emerging science of sensing and manipulation through touch and generally refers to the technology that enables physical interaction with virtual reality environments through force-feedback. The interaction force to the operator is conveyed through a special robotic device referred to as haptic device. Applications span from teaching difficult surgical procedures to medical residents, rehabilitation, psychophysics studies, training of astronauts, virtual CAD (Computer Aided Design), etc.

Several researchers in the haptics community use the term haptic device and the haptic interface interchangeably. Adams et al. [2], however, refers to haptic interface as a link between operator and the virtual reality environment

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that consists of a haptic device and the necessary controller. Henceforth this paper would also refer to the same for the term haptic interface. Moreover, the parameters for the virtual wall impedance would be referred to virtual stiffness and virtual damping only.

Virtual wall forms the basic component or building block of any virtual reality environment. These are typically modeled in software using penalty-based methods that approximate interaction force proportional to stiffness and damping (together referred as impedance) of the virtual wall. With position and velocity input from the haptic interface along with a combination of impedance parameters, a multiple range of interaction forces can be rendered.

Haptic interfaces are often controlled in discrete time, with the continuous time signals at the controller input converted to discrete form, and the discrete time signals at the output converted to continuous form. In the conventional haptics controller, digital implementation of impedance parameters is carried out at uniform sampling rate. Stiffness is typically implemented via position loop, while damping is implemented via velocity loop of the controller. Ideally, a haptic interface should enable a stable interaction with an infinitely stiff virtual wall. However, due to discrete nature of the virtual wall implementation, haptic interfaces tend to be unstable at higher wall impedances. In other words, only a limited combination of the impedance parameters are rendered stably with a particular haptic interface. This limited range is governed by factors such as sampling rate of the controller, sensor resolution, physical damping, and Coulomb friction of the haptic device, etc. [1, 6, 10].

Several researchers established the fact that increasing the sampling rate of the haptics controller would increase stable range of virtual stiffness, [6, 10]. It was also established that virtual damping contributes positively in enhancing overall stiffness and stability of the virtual wall. Besides, the interactions with surfaces like bones, metal-metal contact, etc. would be more realistic for the models with virtual damping, rather than virtual stiffness alone. However, as reported in Colgate and Brown [6], the increase in sampling rate would deteriorate the estimated velocity, which is computed digitally. In this regard, a term *Z-width* was coined to represent a region of virtual stiffness and virtual damping values that a haptic interface could stably implement. It was observed that the desired increase in the sampling rate would reduce the implementation range of virtual damping or diminish the *Z-width* of a haptic interface.

Various methods to improve the *Z-width* of a haptic interface were proposed in [3, 24, 30], with each of them focussing on increasing the physical damping of the device using brakes, electrical damping, and DC motor damping, respectively. Techniques focussing primarily on the sampling rate of the controller were proposed in [5, 14, 18, 23], and [28], resulting in enhancing stability ranges at higher

sampling rates. These techniques, however, utilized uniform sampling rate for the controller, which demanded efficient and accurate velocity estimation, specifically at higher sampling rates, requiring encoders with higher resolution [1, 6], and complex velocity estimation techniques [5, 13].

For a given encoder resolution, improving the stability characteristics of a haptic interface remains a contemporary challenge. In this work, a dual-rate sampling scheme is proposed for the haptics controller in which the position and velocity loops of the controller are decoupled. For stable contact with stiffer walls, the position loop is sampled faster while the velocity loop is sampled at slower rate. The proposition is based on the fact that higher the sampling rates, higher the stable range of the virtual wall stiffness, whereas lower the sampling rate better the estimated velocity. The proposed idea of decoupled sampling draws inspiration from the concept of up- and down-sampling used in the literature of digital control, i.e., multiple sampling rates in different loops/locations of a controller [17].

Note that the idea of multi-rate sampling has been exploited during the last decade in the area of motion control systems [25, 31]. The scheme has been particularly used for the performance enhancement of feed-forward loops in position tracking control and other motion control applications where the basic sampling rate was constrained for one reason or another. Motivation for such multi-rate schemes had been to make the control input smooth when the measurement sampling rate could not be arbitrarily increased due to the sensor bandwidth or any other hardware constraint [25]. Hence, extension of such multi-rate sampling, namely, the dual-rate, to haptics controllers is logical and proposed here. It is to be noted that the idea of dual-rate sampling is different from the multi-rate controllers proposed earlier in the haptics literature [4, 9]. For instance, in the works of [4], multi-rate sampling refers to de-coupling the haptics controller and the virtual environment by implementing a local control loop at higher sampling rate, while simulation (typically involving finite element analyzes) being run at lower sampling rate. The proposed dual-rate scheme concerns the issues with the haptics controller itself whose performance is hampered at higher sampling rates. In the multi-rate scheme proposed by Lee and Lee [18], an additional stiffness control loop at higher sampling rate was added to the conventional haptics controller, with no consideration of the virtual damping.

In this paper, experimental results using a custom-made single-DOF haptic interface employing the conventional and dual-rate sampling schemes are presented first. Theoretical stability analyzes using linear control theory is carried out subsequently to establish the *Z-width* for different sampling rate combinations. Theoretical observations are in agreement with the experimental outcomes, demonstrating the advantages of the proposed scheme in improving the

stable range of virtual wall stiffness, while preserving the implementation range of virtual wall damping.

2 Dual-Rate Haptics Controller

Figure 1 depicts the typical components in an impedance-based haptic interface, also referred as conventional haptics controller in this paper. These controllers are hybrid in nature that consist of both continuous time elements such as human operator $H(s)$, haptic device $D(s)$, as well as, discrete time elements such as environment impedance $E(z)$, unilateral constraint, and sampler T , as shown in Fig. 1. Position data from the optical encoders is sampled and utilized for estimating velocity using finite difference method (FDM) and later for necessary feedback force. The stiffness and damping parameters are implemented using the position and velocity loops, respectively.

In the proposed dual-rate sampling scheme, the position and velocity loops in the haptics controller are sampled at different rates. This is in contrast to the traditional scheme of Fig. 1, where the sampling rates in the haptics controller is kept uniform. In the proposed controller, as shown in Fig. 2, position loop is sampled faster, at T/N , where T is the time period and N represents an integer, as typically used in multi-rate control [17]. Note that the higher sampling rate $f_1 = N/T$ of the position loop improves the stable range of virtual stiffness [6, 10]. On the other hand, the velocity is computed at a lower sampling rate, namely at $f_2 = 1/T$, avoiding exacerbation. For the single-DOF haptic device, these position and velocity information were utilized to calculate the reaction torque τ due to the motion inside the virtual wall. The expression for τ is given by

$$\tau = K\Delta\theta + B\omega \quad (1)$$

where K and B are the virtual stiffness in Nm/rad and the virtual damping in Nms/rad , respectively, whereas $\Delta\theta$, ω are the angular displacement and angular velocity of the

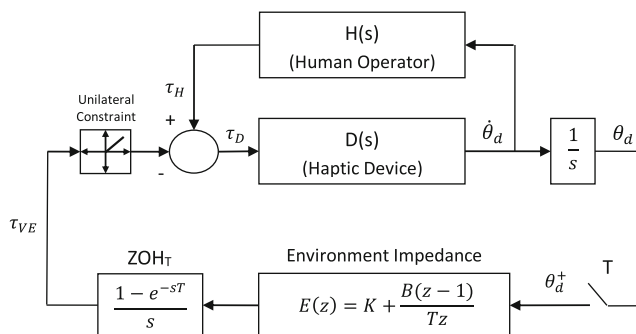


Fig. 1 Conventional haptics controller

haptic device inside the virtual wall, respectively. In a typical haptics application, reaction torque τ outside the virtual wall is usually zero. This was established using a unilateral constraint block shown in Figs. 1 and 2. Equation 1 represents a penalty-based approach of virtual wall interaction with the single DOF rotational haptic device shown in Fig. 3. Equation 1 also represents a linear Kelvin-Voigt model of a visco-elastic surface, i.e., a lumped spring-damper system in mechanical parallel. These models are widely used in haptics applications for their ease of modeling and being computationally simple [6, 7], and [22].

3 Experiments with a Single-DOF Haptic Interface

Virtual wall simulation is a bench-mark experiment typically used for determining the *Z-width* of a haptic interface. Few standard protocols have been followed in the past to obtain the *Z-width* namely, human operated [1, 2, 6], and automated experiments [5, 10, 30]. Both protocols require a uniform velocity or constant external impedance during the trial of wall hit. In the present experiments, a human operator was used as a source of external impedance as he or she plays a fundamental role in typical haptics application. The human operator was trained first about the proper usage of the device, as any arbitrary application of impedance can reach the saturation limit of the actuator. In fact, the operator was thoroughly explained about the onset of instability of the virtual wall. For the experiments, the virtual wall was declared to be unstable if the interaction resulted into constant vibrations/limit cycles along with a shriek sound from the actuator. Hence, only after the human operator was fully familiar with the setup, the actual experiments began.

Figure 3a depicts the components of a single-DOF experimental setup. The device consisted of a 100 mm long link, hinged at one end to achieve rotational motion. The device

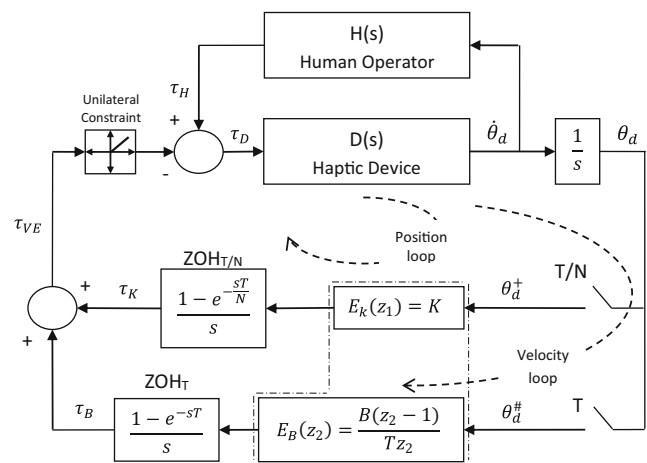
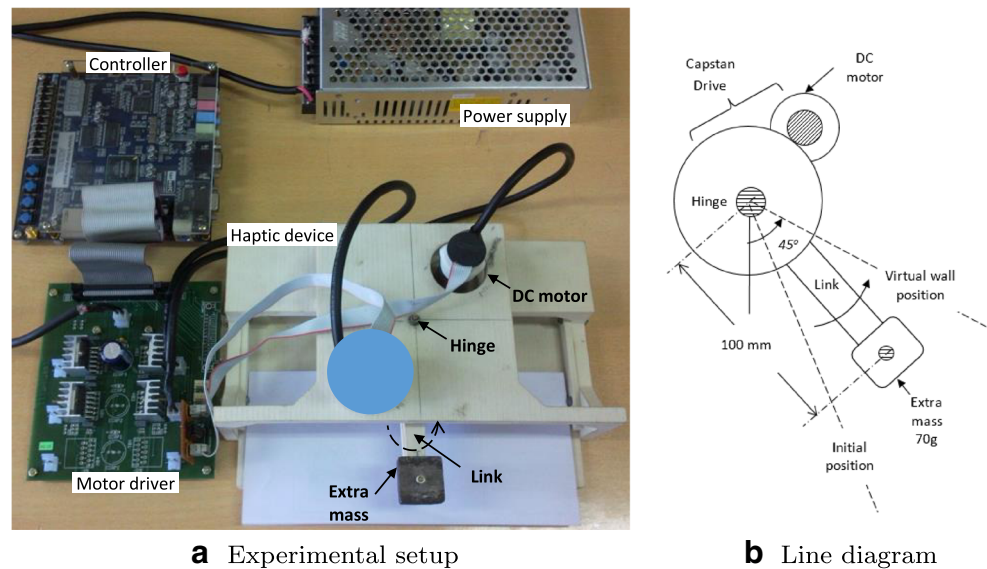


Fig. 2 Proposed dual-rate haptics controller

Fig. 3 Single-DOF haptic device

was driven by a DC motor (Maxon RE-25) with an optical encoder of 1000 counts per revolution mounted. As indicated in Fig. 3b, an amplified torque ($10\times$ motor torque) with a capstan drive mechanism was provided to the link. The free-end of the device was mounted with a 70g mass to increase the impedance during the experiments; similar to the extra mass used in [1, 5]. During the experiments, a virtual wall was defined in the controller at 45° from the horizontal front position. This is indicated in Fig. 3b. A pinch grasp (index finger and thumb) was used by the human operator to rotate the link toward the virtual wall. Actuation commands remained off during the free space motion. As the link penetrated the virtual wall, an opposing torque at the motor axis according to Eq. 1 impeded further motion into the wall. After establishing a stable/unstable contact, the operator pulled back the device from the virtual wall for repeating the trials.

Since the authors were interested in issues at higher sampling rates, all the above experiments were conducted at position loop sampling rate of $f_1 \geq 2\text{kHz}$. In most of the haptic and feedback control applications, velocity computation using FDM is cascaded by a low-pass Butterworth filter for smoothing [5, 6]. An appropriate cut-off frequency (f_c) of about 20% of Nyquist frequency was chosen for the filter, although a smaller fraction could also be chosen. In the dual-rate controller, the velocity was computed from a 2kHz down-sampled position data with $f_c = 200\text{Hz}$ (20% of Nyquist frequency). This specification caused a filter delay of 1ms , which was obtained from impulse response of the filter using MATLAB's filter design and analyzes toolbox (*fdatoool*). Reducing the cut-off frequency below 200Hz would sometimes provide a better velocity estimate depending upon the system characteristics. However, the authors restricted to the above specification in order to avoid further

increase in the filter delay. For example, with $f_c = 30\text{Hz}$, the delay would increase to 10ms for the sampled data of 2kHz. Increased delay in the control loop significantly hampers the overall stability of the haptic interface and hence is undesirable for rendering stiff virtual walls. Note that the control inputs to the actuator were fed at higher sampling rates than that associated with the velocity loop. The velocity for the intervening period was assumed to be equal to the last measured velocity. One could think of some prediction/estimation techniques to estimate velocity in between the samples also. However, that would be needed only when the velocity changes are expected to be at a very high rate which is rarely encountered in haptics applications.

Now, in order to determine the experimental *Z-width* of the haptic interface, a combination of K and B values were set in the controller first, followed by the virtual wall experiments to determine the nature of interaction. An initial experimental study in [15] revealed that the maximum stable value of virtual stiffness K without virtual damping (i.e., $B = 0$) would saturate for the sampling rates above $f_1 = 20\text{kHz}$. In other words, increasing the sampling rate beyond 20kHz would lead to only insignificant improvements in the stable range of K . Therefore all experiments in this study were carried out below the sampling rate of 20kHz. The same was considered during the theoretical studies too. It must be noted that [15] provided the detailed experimental procedure with several results and implications that have not been repeated here. In the present paper, a more refined experimental *Z-width* was obtained after proper identification of the physical parameters of the haptic device. In addition, the present experimental results achieved good repeatability. Moreover, the present paper provides the detailed theoretical analyzes of the controller, as shown in Section 5, which was not reported in [15].

Now, for the uniform rate scheme, as similar to the results of Colgate et al. [6], a decrease in the experimental *Z-width* was observed after increasing the sampling rate from 2kHz to 4kHz, as shown in Fig. 4. Increased sampling rate of the position loop initially corresponded to increase in the value of K from 125 to 200 Nm/rad , as seen in Fig. 4 at $B = 0$. However, as the sampling rate increased from 2kHz to 4kHz, a decrease in the maximum implementation range of B was observed, i.e., B decreased from 0.9 to 0.65 Nms/rad . For the case of sampling rates above 4kHz, i.e., at $f_1 = f_2 = 10kHz$ and 20kHz, the stable value of K similarly increased at $B = 0$, i.e., to 320 and 354 Nm/rad , respectively. However, due to considerable noise in the estimated velocity even after appropriate filtering, the implementation range of virtual damping suffered considerably. In other words, the virtual damping part of the haptic force law in Eq. 1 could not contribute in increasing the overall stiffness of the virtual wall at 10kHz and 20kHz, respectively. These observations represent a common trend [6] in the *Z-width* of a haptic interface at higher sampling rates.

For the experimental *Z-width* using the proposed dual-rate scheme, sampling rate of the position loop was varied as $f_1 = Nf_2$, for $N = 1, 2, 5$ and 10, with $f_2 = 2kHz$. Experiments were conducted by increasing the value of K in fixed steps for a particular value of B in the controller. Using the criteria for instability, maximum stable value of K was noted for a particular value of B and the sampling rate combination. Ten repeated trials were conducted at each combination of B and f_1 to assess the average value of the stable virtual stiffness. Such repeated trials are a common practice in virtual wall experiments, as human impedance varies with time [12]. Figure 5 depicts the *Z-width* of the haptic interface for the following sampling rate combinations: (a) $f_1, f_2 = 2kHz$, denoted as 2-2; (b) $f_1 = 4kHz, f_2 = 2kHz$ denoted as 4-2; (c) $f_1 = 10kHz, f_2 = 2kHz$

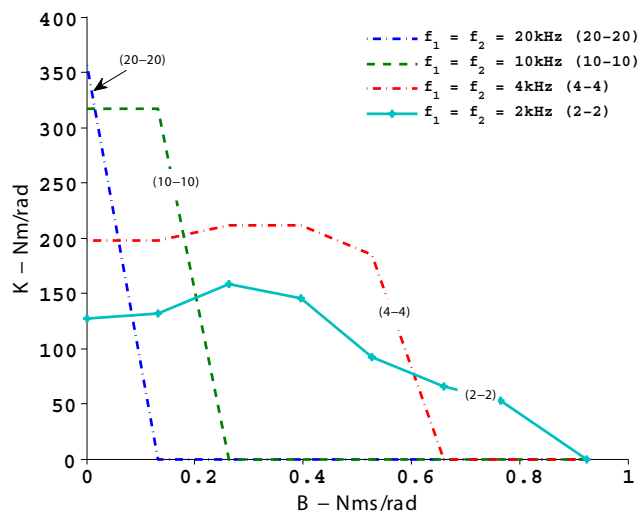


Fig. 4 Experimental *Z-width* with uniform rate scheme

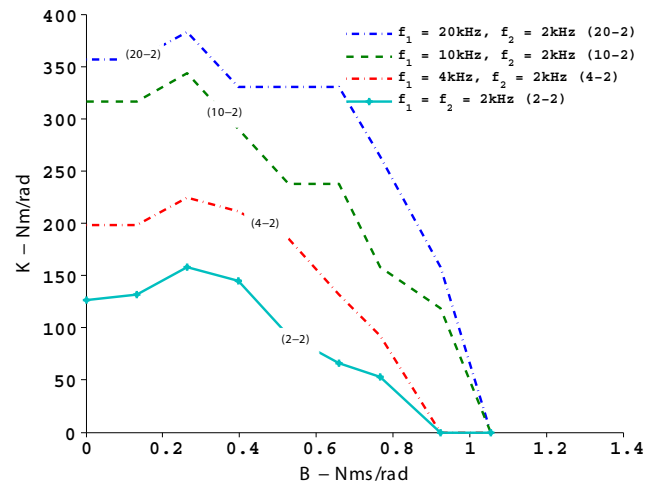


Fig. 5 Experimental *Z-width* with dual-rate scheme

denoted as 10-2, and (d) $f_1 = 20kHz$ and $f_2 = 2kHz$ denoted as 20-2. As seen from Fig. 5, for all the combinations, i.e., 2-2, 4-2, 10-2 and 20-2, an improvement in the stable value of virtual stiffness was achieved for virtual damping ranging from $B = 0$ to 0.2 Nms/rad . This is in accordance to the fact established in [6, 10] that virtual damping contributes positively to the stability of a haptic interface up to a particular limit. For the combinations, 4-2 and 2-2, as the value of B was increased further, the corresponding value of K dropped until $B = 0.9$ Nms/rad . For $B > 0.9$ Nms/rad , virtual wall was unstable for even very low range of virtual stiffness. Similarly, for combinations of 10-2 and 20-2, the value of K dropped for $B > 0.3$ Nms/rad until $B = 1.05$ Nms/rad . As can be clearly seen from these plots, the value of stable K was quite high for combination of 20-2 compared to the one in 2-2. Additionally, the maximum stable range of virtual damping was preserved in the combination of 20-2 rather than decreased on account of higher sampling rate of the velocity loop using the conventional controller. Since the velocity in the proposed scheme 20-2 was estimated at $f_2 = 2kHz$, the implementation range of B was similar to the one with 2-2 scheme. Also, the maximum achievable virtual stiffness for 20-2 combination at $B = 0.8$ Nms/rad is more than 200 Nm/rad , which is higher than the one with 2-2 and others. In fact, the maximum virtual stiffness follows the trend of position loop sampling, i.e., $B = 0$ to 0.8 Nms/rad , whereas K varies according to 20-2 > 10-2 > 4-2 > 2-2.

The above experimental outcomes clearly establish the advantages of the dual-rate haptics controller over the conventional one in realizing larger values of stable impedances at higher sampling rates. It must be pointed out here that the above experiments may incur errors from various sources, which were not accounted for in this work. For instance, non-linear effects like actuator saturation, device inertia,

variations in human operator impedance, etc. have a minor impact in modifying the stable limits of impedance parameters. Accounting for such effects is beyond the scope of this study, but will be considered in future.

4 Stability Analyzes

Since a haptic interface involves human operator in the loop, ensuring stability of such system is vital. Theoretical study of such systems enables one to identify bounds on various controller parameters, guaranteeing a stable virtual wall interaction in practice. An improvement in performance or vice-versa with different controller schemes can also be studied while complementing the experimental observations to some extent.

In this section, theoretical insights into the effect of dual-rate sampling on the stability of a haptic interface are provided. Several researchers have utilized linear control theory to arrive at conservative bounds on the controller parameters rather than an exact match between the experimental and theoretical observations [10, 19, 22]. The authors would like to complement the improved trend in the experimental *Z-width* reported in Section 3.

A haptics controller consists of both continuous time as well as discrete time components in the loop. The device (also referred to as plant for the controller) represents the continuous time component, while the virtual wall is the digital part. These hybrid systems are theoretically analyzed by either solving them completely in digital domain or in the continuous time domain. To perform the stability analyzes in continuous time domain, the digital part of the controller is transformed to continuous domain first. Later, the system is solved using the well-established continuous time domain techniques [18]. Gill et al. [10], however, discretized the plant first, and obtained the characteristic equations of the closed-loop system in discrete time domain. Using bilinear/Tustin transformation, the characteristic equation was then converted to continuous time domain so that the Routh's criteria could be applied.

As a first step toward the analyzes, all discrete time components of the haptic interface were converted into continuous time domain. For that, bilinear transformation, which is popularly utilized for such conversions was used. It represents an exact mapping of the system in discrete time domain to continuous time domain, and is given by

$$z = \frac{1 + \frac{T_s}{2}}{1 - \frac{T_s}{2}} \quad (2)$$

where z is used to represent a system in discrete time domain, similar to the use of s for continuous time domain representation. In the following analyzes, it is assumed that the position and velocity loops of the proposed controller

shown in Fig. 2 are sampled at T/N and T periods, respectively. For analyzes purpose, this multiple sampling within the controller requires the dynamics of each loop be represented with a unique z variable. For instance, z_1 and z_2 are used here to represent the discrete time variables associated with the position and velocity loops, respectively. Now, using the bilinear transformation of Eq. 2, each discrete time variable can be represented as

$$z_1 = \frac{1 + \frac{T_s}{2N}}{1 - \frac{T_s}{2N}}, \quad z_2 = \frac{1 + \frac{T_s}{2}}{1 - \frac{T_s}{2}} \quad (3)$$

For the dual-rate haptics controller, each sampler introduces a hold effect which is popularly estimated using the zero-order-hold (ZOH) function given by

$$ZOH = \frac{1 - e^{-Ts}}{s} \quad (4)$$

Accordingly, the ZOH functions associated with the position and velocity loops, denoted as ZOH_K and ZOH_B , respectively, are expressed as follows

$$ZOH_K = \frac{1 - e^{-\frac{T_s}{N}}}{s}, \quad ZOH_B = \frac{1 - e^{-Ts}}{s} \quad (5)$$

Note that, as per notation of ZOH_T used in Fig. 1 and ZOH_{T1} or ZOH_{T2} used in Fig. 2, $ZOH_K \equiv ZOH_{T/N}$ and $ZOH_B \equiv ZOH_T$. The new notations are introduced for the sake of brevity in the ensuing derivations only. Substituting, exponential function in ZOH_K with its equivalent Taylor series expression, one has

$$\begin{aligned} ZOH_K &\cong \frac{1 - (1 - \frac{T_s}{N} + \frac{T_s^2}{2N^2})}{s} \\ &= \frac{T}{2N} (2 - \frac{T_s}{N}) \end{aligned} \quad (6)$$

where 3^{rd} and higher-order terms of the Taylor series are neglected due to insignificant contribution to the ZOH function. Similarly, for the velocity loop one can write

$$ZOH_B = \frac{T}{2} (2 - Ts) \quad (7)$$

Equations 6 and 7 represent linear approximations of the ZOH functions associated with the position and velocity loops, respectively. With the above approximations in place now, the transfer function of the position loop in Fig. 2 can be written as

$$\begin{aligned} \frac{\tau_K}{\theta_d} &= ZOH_K \cdot K \cdot S_K \\ &= \frac{K}{2} (2 - \frac{sT}{N}) \end{aligned} \quad (8)$$

where S_K acts as transfer function of the sampler in the position loop given by $\frac{N}{T}$ [18].

In the next step, transfer function of the velocity loop in Fig. 2 was obtained. For detailed insights into the analysis,

a second-order Butter-worth filter was also included while writing the transfer function of velocity loop as

$$\frac{\tau_B}{\theta_d} = ZOH_B \cdot B \cdot \frac{z_2 - 1}{Tz_2} \cdot S_B \cdot U(z_2) \quad (9)$$

where $\frac{z_2 - 1}{Tz_2}$ represents the discrete time transfer function of the first-order differentiator to estimate velocity using FDM, while S_B acts as transfer function of the sampler in the velocity loop given by $\frac{1}{T}$, and $U(z)$ denotes the transfer function of the second-order Butter-worth filter in z -domain. Using Eq. 3, components on the right hand side (RHS) of Eq. 9 were transformed into continuous domain variables as follows:

Continuous time equivalent of the first-order differentiator is given by

$$\frac{z_2 - 1}{Tz_2} = \frac{2s}{2 + Ts} \quad (10)$$

Equation 10 is also referred to as transfer function of a practical differentiator in continuous time domain. This is in contrast to an ideal differentiator whose transfer function is given by 's', which cannot be realized practically due to its unstable nature [21].

Next, the second-order Butter-worth filter $U(z_2)$ is represented in z -domain [11] as

$$U(z_2) = \frac{d_1 + d_2 z_2^{-1} + d_3 z_2^{-2}}{1 + c_1 z_2^{-1} + c_2 z_2^{-2}} \quad (11)$$

where d_1, d_2, d_3 and c_1, c_2 are the necessary filter coefficients which are functions of filter-order and cut-off frequency ' f_c '. Using the *fdtool* of MATLAB, the filter coefficients with $f_c = 200\text{Hz}$ were obtained and substituted in Eq. 11. The continuous domain equivalent of the above filter is written as

$$U(s) = \frac{4\tilde{d}_1 + 4\tilde{d}_2 Ts + \tilde{d}_3 T^2 s^2}{4\tilde{c}_1 + 4\tilde{c}_2 Ts + \tilde{c}_3 T^2 s^2} \quad (12)$$

where $\tilde{d}_1 \equiv d_1 + d_2 + d_3$; $\tilde{d}_2 \equiv d_1 - d_3$; and $\tilde{d}_3 \equiv d_1 - d_2 + d_3$; while $\tilde{c}_1 \equiv 1 + c_1 + c_2$; $\tilde{c}_2 \equiv 1 - c_2$; and $\tilde{c}_3 \equiv 1 - c_1 + c_2$. Transfer function for the velocity loop in the s -domain is thus obtained by substituting the expressions of ZOH_B , Eq. 7, the differentiator, Eq. 10, and the filter $U(s)$, Eq. 12, into Eq. 9 as

$$\frac{\tau_B}{\theta_d} = \frac{(2Bs - BTs^2)(4\tilde{d}_1 + 4\tilde{d}_2 Ts + \tilde{d}_3 T^2 s^2)}{(2 + Ts)(4\tilde{c}_1 + 4\tilde{c}_2 Ts + \tilde{c}_3 T^2 s^2)} \quad (13)$$

Using Eqs. 8 and 13, the transfer function of the feedback block in Fig. 2 including filter, represented for simplicity as $E(s)$, is given by

$$E(s) \equiv \frac{\tau_k}{\theta_d} + \frac{\tau_B}{\theta_d} \quad (14)$$

In the next step, transfer function of the controller plant (i.e., human operator and the haptic device) in Fig. 2 is obtained as:

$$P(s) = \frac{D(s)}{1 + D(s)H(s)} \quad (15)$$

where $D(s)$ represents the transfer function of the single-DOF haptic device at hand, which is given by $D(s) = \frac{1}{Is^2 + bs}$, in which I and b are the inertia and physical damping of the device. The term $H(s)$ on the other hand represents the transfer function of the human operator.

In [20], impedance of a human arm was modeled as a linear second-order system having some mass, damping and stiffness that results into most conservative bounds on the stable virtual wall parameters, as shown in Gill et al. [10]. The same is adopted here. Accordingly, the transfer function of the human operator is expressed as

$$H(s) = M_h s^2 + B_h s + K_h \quad (16)$$

where M_h, B_h and K_h are the nominal inertia, viscous damping and stiffness coefficients of the human arm, respectively. The corresponding values in [20] are transformed to rotational domain (I_h, b_h and k_h) for conformity with our theoretical analyzes of the rotational single-DOF system.

Substituting the expressions of $D(s)$ and $H(s)$ from above, the transfer function of the controller plant in Eq. 15 reduces to

$$P(s) = \frac{\frac{1}{Is^2 + bs}}{1 + \frac{1}{Is^2 + bs}(I_h s^2 + b_h s + k_h)} \quad (17)$$

$$= \frac{1}{(I + I_h)s^2 + (b + b_h)s + k_h}$$

The closed-loop transfer function for the haptics controller including filter and human impedance, represented as $G(s)$, is finally obtained as

$$G(s) = \frac{P(s)}{1 + P(s)E(s)} \quad (18)$$

$$= \frac{g_3 s^3 + g_2 s^2 + g_1 s + g_0}{h_5 s^5 + h_4 s^4 + h_3 s^3 + h_2 s^2 + h_1 s + h_0}$$

where the coefficients g_i 's and h_i 's of Eq. 18 are provided in Appendix A.1. From the denominator of $G(s)$ in Eq. 18, the characteristic equation for the haptics controller is given by

$$h_5 s^5 + h_4 s^4 + h_3 s^3 + h_2 s^2 + h_1 s + h_0 = 0 \quad (19)$$

The inertia ($I = 0.000929 \text{ kgm}^2$) and friction ($b = 0.0264 \text{ Nms/rad}$) parameters of the haptic device were estimated using an identification procedures similar to the one followed in Abbott and Okamura [1]. For stability analyzes, Routh's criteria was applied to obtain the magnitude of the stable impedance parameters from Eq. 19, for various combinations of T and N .

Figure 6 depicts the stable range of impedance parameters or Z -width of the 1-DOF haptic interface at hand

including the effects of human operator impedance and velocity filter for various sampling rate combinations. As evident from Fig. 6, the stable range of virtual stiffness improved with the increase in the sampling rate of the position loop, while the range of the virtual damping remained constant. This is in agreement with the experimental observations of Fig. 5. However, the values of the impedance parameters are on a higher range, which are expected due to the unaccounted elements of the haptics control loop. These are discussed in Section 5. Theoretical results in Fig. 6 confirm the advantages of the proposed dual-rate approach in realizing an improved *Z-width* of the haptic interface.

5 Discussion

This paper proposes a modified controller strategy for improving the performance of impedance based haptics interface at high sampling rates. Stability analyzes presented in Section 4 included the effect of human operator impedance and the Butter-worth filter. It must be noted that, an actual haptics controller contains several other non-linear components viz. unilateral constraint block, encoder quantization, Coulomb friction, etc. Besides, other complex-to-model effects may creep in during the virtual wall experiments, namely, motor saturation, amplifier bandwidth [1], thermal effects, vibration modes of the device structure [8], etc. Such effects have only caused deviation in the experimental and the theoretical *Z-width* of the haptic interface.

5.1 Simulink Results

In order to obtain *Z-width* closer to the experimental results, a Simulink model of the haptics controller was developed to simulate the haptics interface as realistically as possible. Similar to the procedure explained in Koul et al. [16], the 1-DOF haptics controller with dual-rate sampling was

developed in Simulink environment of MATLAB. Note that MATLAB is a simulation tool which has been successfully used in many domains, e.g., robotics [26], meteorology [27], agriculture [29], etc.

Several non-linear elements viz. Coulomb friction, unilateral constraint, sampler, and encoder quantization were appropriately incorporated in the model. An automated protocol was incorporated to carry out *Z-width* simulations for several sampling rate combinations. Automated protocol was opted owing to the difficulty of modeling the impedance of an human operator exactly [10].

Figure 7 depicts simulated *Z-width* of the 1-DOF haptics interface with the proposed controller. The above results demonstrate nearly similar improvements in the stable range of stiffness while preserving the stable range of damping, as observed in Fig. 5. A comparison of the simulated and experimental *Z-width* is elucidated in Fig. 8. As evident from Fig. 8, Simulink model of the dual-rate haptics controller provides better match to the experimental results. Further improvements in matching simulated and experimental outcomes may be possible by incorporating several other complex-to-model elements, which is beyond the scope of this work.

With regards to the experimental observations, the maximum stable range of virtual damping was preserved even after increasing the sampling rate of the position loop. This is quite advantageous, especially for applications that require stiffer wall interaction over a good range of virtual damping. As an example, using the proposed dual-rate scheme, one can achieve stable interaction with the virtual wall even with the following specifications: $B = 0.8$ Nms/rad, $K > 200$ Nm/rad (for $f_2 = 2$ kHz only). Increasing f_1 beyond 2 kHz does not thereby hamper the process of velocity estimation, as evident from the theoretical as well as experimental observations. A comparison of the experimental *Z-width* with uniform rate and dual-rate sampling is provided in Fig. 9.

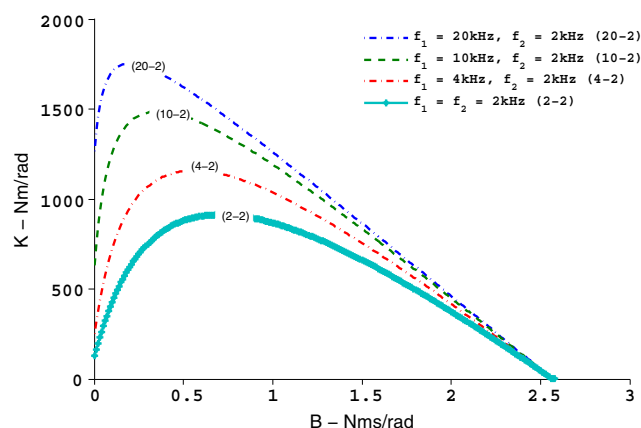


Fig. 6 Theoretical *Z-width* including filter and human impedance

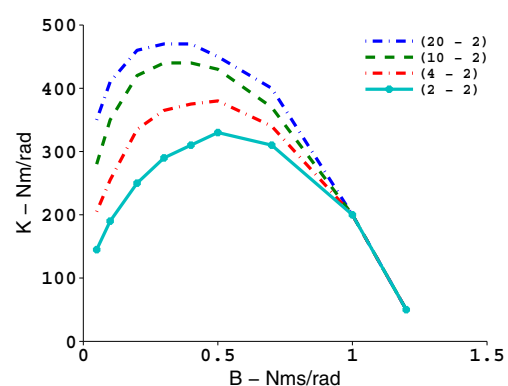
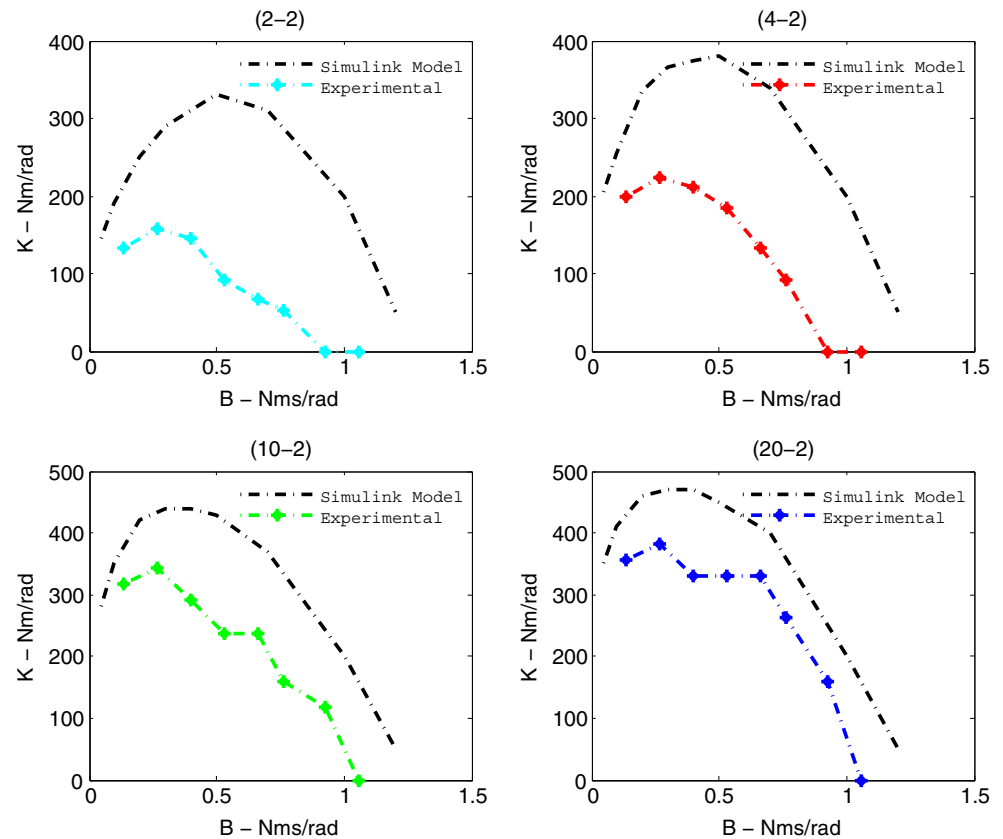


Fig. 7 Simulink based *Z-width*

Fig. 8 Comparison of Z -width

5.2 Stability Criteria

An interesting observation during the theoretical analyzes of dual-rate haptics controller led the authors to a new set of stability criteria for a small range of virtual damping. For instance, as proved in Appendix A.2 (a specific case), the discrete differentiator function $\frac{z^2-1}{Tz^2}$, if approximated by an ideal differentiator whose transfer function is given by 's', leads to the following stability criteria

$$K < \frac{2(b+B)N}{T}, \quad B < \frac{2I}{T} \quad (20)$$

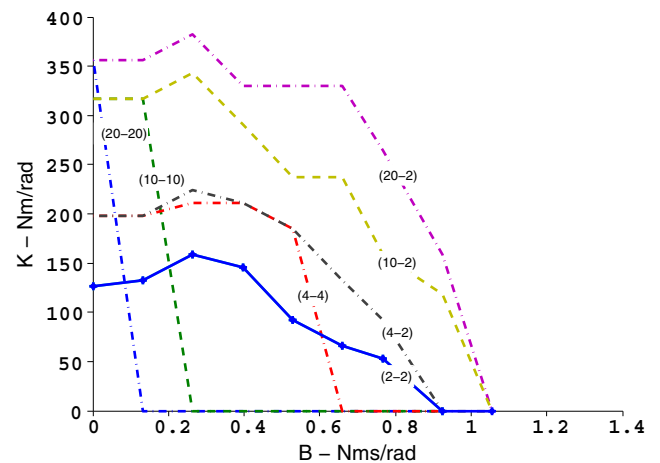
If $N = 1$, Eq. 20 reduces to the well known stability criterion of Minsky et al. [22], and Gil et al. [10] for the uniform rate haptics controller, i.e.,

$$K < \frac{2(b+B)}{T} \quad (21)$$

Although Eq. 20 is valid only for a small range of virtual damping and sampling-rate of the controller, it implies two major facts: firstly, increase of the sampling-rate ratio N , would increase the stable range of the virtual wall stiffness, and secondly, the device inertia I significantly impacts the implementation of virtual damping. In fact, the device inertia along with the sampling rate of the velocity loop bounds the stable range of the coefficient of the virtual damping.

5.3 Automated experiments

The advantages of the proposed 20-2 combination over the uniform-rate sampling of 20-20 combination were also demonstrated using automated wall hit experiments. The actuator of the haptics interface was commanded with a torque trajectory outside the virtual wall so that the end-effector (link) would hit the wall with uniform velocity. After hitting the virtual wall, a reaction torque in the opposite direction (commanded as per the haptics force law in

**Fig. 9** Comparison of experimental Z -width

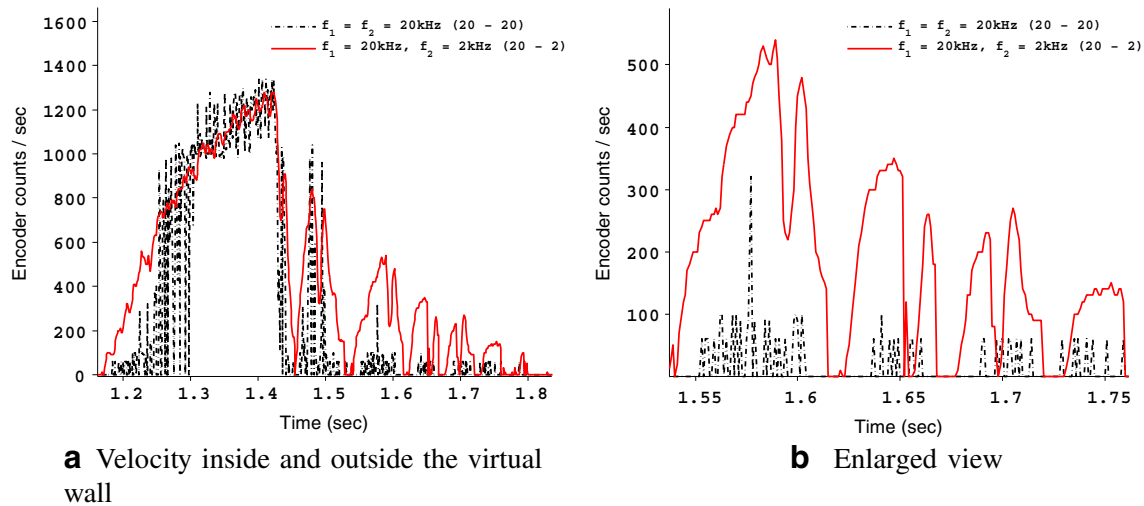


Fig. 10 Comparison of velocity during automated wall hit trials

Eq. 1) resisted forced motion of the end-effector into the virtual wall. The combined effect of the commanded and the reaction torque outside and inside the virtual wall, respectively, resulted into momentarily oscillations near the virtual wall till the system came to rest. In the experiments, the virtual wall was hit at around 1.41 sec, while the system came to rest (steady state) at around 1.8 sec.

The angular velocity profile of the actuator and the depth of penetration inside the virtual wall were compared during the trials. The depth of penetration (of the end-effector) inside the virtual wall is directly dependent on the impedance of the wall. A wall offering higher impedance would allow lesser penetration, while a lower impedance wall would allow comparatively higher penetration. Figure 10 depicts the corresponding values of angular velocity of the actuator during one of the trials. Angular velocity is directly proportional to the difference in encoder counts

per unit time (as per FDM of velocity estimation). As can be clearly seen, considerable noise is associated with the estimated angular velocity even after filtering in the conventional scheme. On the contrary, angular velocity estimation is much better in the dual-rate scheme. The same is true before hitting the virtual wall (till 1.4 seconds) and after oscillating near the virtual wall (from 1.45 to 1.75 seconds). Thereby, the estimation is better in the proposed scheme even at low velocity after hitting the virtual wall.

Similarly, a comparison of the depth of penetration for the same trials is presented in Fig. 11a. The same was measured as the amount of distance traveled (penetration) inside the virtual wall (in terms of encoder counts). Lower depth of penetration with 20-2 combination compared to the 20-20 case in Fig. 11b further confirms the effective utilization of the virtual wall damping. For 20-2 combination, the overall impedance of the virtual wall increased. This is certainly

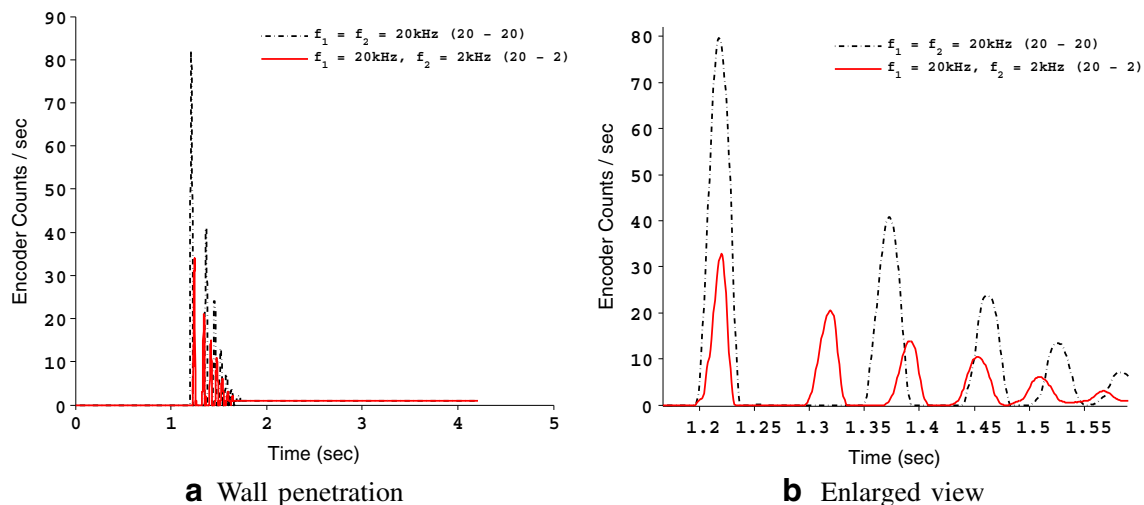


Fig. 11 Comparison of depth of penetration during automated wall hit trials

due to the better velocity estimation in the proposed scheme.

Toward the end, it is reiterated that the techniques presented in haptics literature assume uniform sampling of both the position and velocity loops, thereby, requiring fancier and complex velocity estimation procedures to improve the otherwise diminished *Z-width* at higher sampling rates. The technique presented in this work avoids the use of such complex estimation procedures by modifying the basic sampling scheme of the haptics controller. Simple velocity estimation procedure viz. finite difference of the position over a short period of time was considered. Similar improvements in the *Z-width* may also be obtained using sophisticated encoders and complex estimation tools. However, augmenting the proposed scheme with any fancier velocity estimation technique would hamper the performance of the haptics controllers due to the increased delay in the estimation of velocity with several methods in the loop. Hence either of the approaches, i.e., proposed dual-rate or the sophisticated velocity estimations could be used to enlarge the *Z-width* of a haptics controller.

6 Conclusions

In this work, how a dual-rate haptics controller can be utilized to enhance the stable range of virtual wall stiffness at higher sampling rates is demonstrated. The proposed de-coupled sampling safely allowed one to increase the sampling rate of the position loop without worrying about the consequences on the estimated velocity. Appropriate sampling of the velocity loop enabled smoother estimation, which otherwise worsened at higher rates. Experimental outcomes clearly demonstrated the effectiveness of the proposed scheme by successfully implementing the virtual damping over a similar range to that of lower sampling rate.

The stable range of virtual stiffness improved at each value of virtual damping with the dual-rate scheme. Theoretical analyzes also confirmed the improved stability characteristics with the dual-rate scheme. Besides, a new set of general stability conditions, namely, Eq. 20 were derived which explained the well-known existing conditions.

The conventional uniform-rate scheme requires a complex velocity estimation technique, along-with higher resolution encoders, for preserving the stable range of virtual damping. The proposed scheme avoids all such sophisticated and expensive utilities.

Hence, the paper contributes to the development of new knowledge, namely, the improved range of virtual stiffness at higher sampling rates, and new stability conditions based on the dual-rate sampling and theoretical stability analyzes, respectively. As a continuation of this work, the proposed dual-rate scheme was implemented on a 2-DOF haptic interface, which will be reported separately.

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Appendix A

A.1 Coefficients in the Stability Analyzes

Details of the various coefficients in the stability analyzes in Section 4 are as follows:

$$\begin{aligned} g_3 &= 2N\tilde{c}_3T^3; g_2 = 8N\tilde{c}_2T^2 + 4N\tilde{c}_3T^2; g_1 = 16N\tilde{c}_2T + 8N\tilde{c}_1T; g_0 = 16N\tilde{c}_1; \\ h_5 &= \tilde{I}g_3; h_4 = \tilde{I}g_2 + \tilde{b}g_3 - K\tilde{c}_3T^4 - 2BN\tilde{d}_3T^3; \\ h_3 &= \tilde{I}g_1 + \tilde{b}g_2 + k_hg_3 - 4K\tilde{c}_2T^3 + 2K(N-1)\tilde{c}_3T^3 + 4BN\tilde{d}_3T^2 - 8BN\tilde{d}_2T^2; \\ h_2 &= \tilde{I}g_0 + \tilde{b}g_1 + k_hg_2 + 4KN\tilde{c}_3T^2 + 8K(N-1)\tilde{c}_2T^2 - 4K\tilde{c}_1T^2 + 16BN\tilde{d}_2T - 8BN\tilde{d}_1T; \\ h_1 &= \tilde{b}g_0 + k_hg_1 + 16KN\tilde{c}_2T + 8K(N-1)\tilde{c}_1T + 16BN\tilde{d}_1; h_0 = k_hg_0 + 16KN\tilde{c}_1; \end{aligned} \quad (22)$$

where $\tilde{I} \equiv I + I_h$, and $\tilde{b} \equiv b + b_h$.

A.2 Stability Analyzes using an Ideal Differentiator

In this section, some insights into the theoretical stability analyzes of the dual-rate controller are provided without considering the effects of the human operator impedance and the velocity filter. The analyzes are specifically focussed on

substitution of discrete differentiator function $\frac{z^2-1}{Tz_2}$ in Eq. 9 with an ideal derivative whose transfer function is ‘s’.

With the above assumption, Eq. 9 in s-domain reduces to

$$\begin{aligned} \frac{\tau_B}{\theta_d} &= \frac{T}{2}(2 - Ts) \cdot B \cdot s \cdot \frac{1}{T} \\ &= \frac{B}{2}(2s - Ts^2) \end{aligned} \quad (23)$$

The transfer function, $E(s)$ of Eq. 14, can then be rewritten as

$$E(s) \equiv \frac{K}{2N}(2N - Ts) + \frac{B}{2}(2s - Ts^2) \quad (24)$$

The closed-loop transfer function $G(s)$ is given next as

$$\begin{aligned} G(s) &= \frac{D(s)}{1 + D(s)E(s)} \\ &= \frac{2N}{(2NI - BNT)s^2 + (2Nb - KT)\dots \\ &\quad \dots + 2NB)s + 2NK} \end{aligned} \quad (25)$$

Routh's stability criteria requires that roots of the denominator in Eq. 25 be positive. In other words, the following stability criteria is obtained:

$$K < \frac{2(b + B)N}{T}, \quad B < \frac{2I}{T} \quad (26)$$

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